FOR COMPLETE BEGINNERS

Java printed 21. Why?

Complete Modern Java, episode 6: values and types in Java and TypeScript

> Every value has a type, and the type decides what happens to it.

Two loaves and one milk printed Items: 21. The values were right; the types decided what + did.

Value, variable, type, literal, assignment, then the everyday types in both languages.

In this episode

- Value, variable, type, literal
- int, double, boolean, String, char
- The compiler checking types
- TypeScript number, boolean, string

Coming next

- Expressions and operators (episode 7)
- Numbers and precision (episode 8)
- Decisions (episode 9)

At the time of recording: Java 25, Node 24 LTS, IntelliJ IDEA 2026.1.3.

01 What each line printed

"Items: " + bread + milk	`Items: 21`
"Items: " + (bread + milk)	`Items: 3`
double whole = 4;	`4.0`
int price = 1.5;	`error: possible lossy conversion from double to int`
let quantity: number = "three"	`TS2322: Type 'string' is not assignable to type 'number'`
let inferred = 3; inferred = "three"	`TS2322 as well: inferred types are checked`

Every row was run, not predicted.

02 Four words, one line

int bread = 2; : 2 is the value (a literal), bread the variable, int the type.

= is assignment: it puts 2 into bread. It does not ask whether they are equal.

Java types

```
int quantity = 3;          // whole numbers
double price = 4.99;       // decimal point
boolean paid = false;      // true or false
String product = "Milk";   // text, double quotes
char grade = 'A';          // one character, single quotes
```

> Printed 3, 4.99, false, Milk, A.
>
> Run on a Mac, 2026-10-23: Java 25.0.4.1, Node 24.22.1, TypeScript 7.0.2.

03 TypeScript

One number type for whole numbers and decimals; string with a small s; inference is still checked.

types.ts

```
let quantity: number = 3;
let price: number = 4.99;
let paid: boolean = false;
let product: string = "Milk";
let inferred = 3;           // a number, by inference
```

> npx tsc --noEmit was quiet; node printed 3 4.99 false Milk 3.
>
> Run on a Mac, 2026-10-23: Java 25.0.4.1, Node 24.22.1, TypeScript 7.0.2.

Pick the type

1. Counting? int.
2. Measuring, with a decimal point? double.
3. Yes or no? boolean.
4. Text? String in Java, string in TypeScript.
5. Challenge: predict three lines before you run them.

Everything else

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The code and the challenge files: complete-modern-java
https://github.com/code-with-sam-dev/complete-modern-java

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